

Integrate Engineering Design



DAEN 640 Capstone Senior Design

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Updates

BH: reach out for data access

COE: data/modelling/prototyping

ML: simulation models/data generation

Next week: PM, revised goals, Questions

Check Canvas

Plan for today

Engineering Design Process

Integrated Engineering Design Process

System Architecture, Data Pipeline

The Toaster Project

Challenges, Solutions

Learning Goals

Map the V-Model

Identify Configuration Items

Understand Traceability

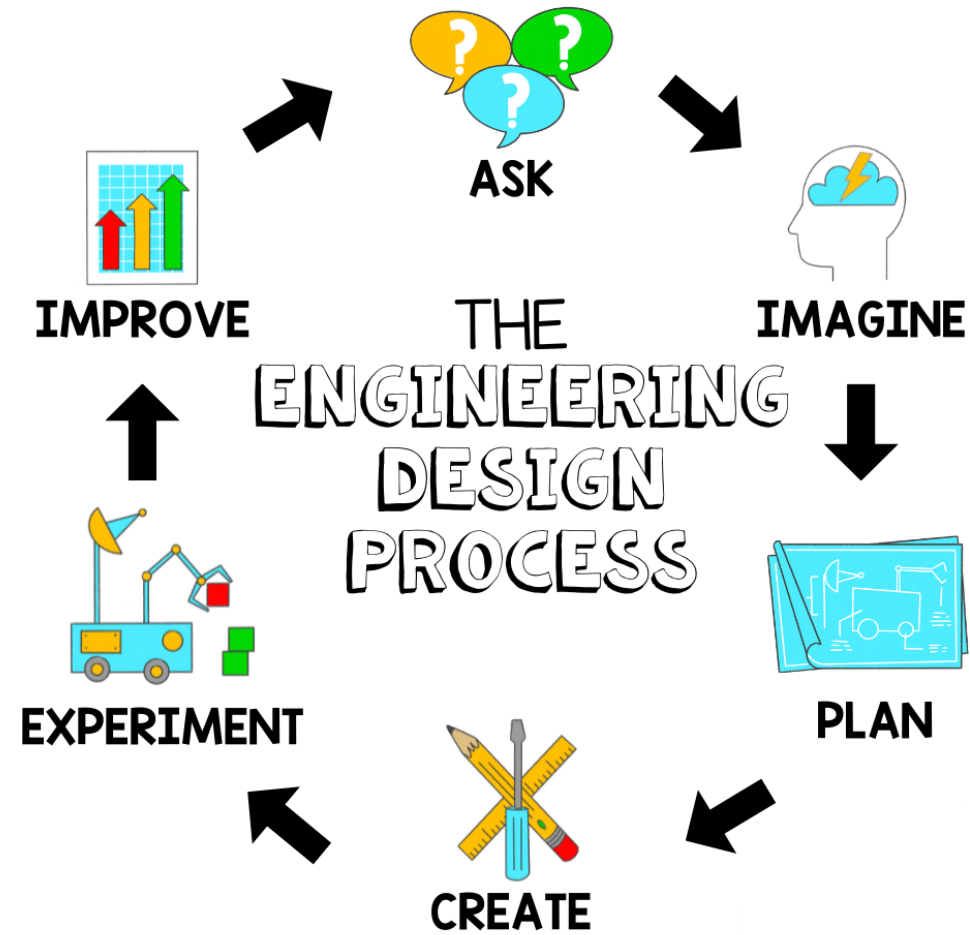
Connect Tests to Artifacts

Evaluate Change Impact

Descriptions

- **Map the V-Model**
See how requirements, design, and implementation connect to testing and validation.
- **Identify Configuration Items**
Recognize key items (requirements, schemas, code, pipelines, infrastructure) and how they are versioned.
- **Understand Traceability**
Follow how requirements link to system components, data pipelines, and tests.
- **Connect Tests to Artifacts**
Identify what is tested or validated for each major configuration item.
- **Evaluate Change Impact**
Understand how changes ripple across design, implementation, and testing.

Engineering Design Process



A step-by-step approach engineers use to identify problems, brainstorm solutions, build and test prototypes, and improve designs to create effective, practical solutions.

Engineering Design Process

Similarities to the scientific method

- Begin with existing knowledge, gradually become more specific

Key difference?

- The scientific process emphasizes explanation, prediction, and discovery.
- The engineering process focuses on application of systematic thinking to leverage technology in service of human needs!



<https://www.youtube.com/watch?v=qq3fxfmWr4>

Life Lessons from a Pencil

What matters is what's inside

Keep yourself sharp

Mistakes are part of the process

Pressure creates output

Collaboration matters

Durability comes from design

Imagine a pencil journey as a metaphor

Raw Materials → Shaping → Testing → Final Pencil

Engineering Design Process

1. Define the problem (*Why make a pencil?*)
2. Research & brainstorm (*What materials & designs?*)
3. Develop concepts (*Sketch ideas, pick the best*)
4. Build & prototype (*Make a sample pencil*)
5. Test & evaluate (*Does it write? Break?*)
6. Redesign/refine (*Improve core, casing, process*)

Engineering Design Process

**Material Selection
and Functionality**

**Mapping Process to
Design Stages**

**Balancing
Constraints**

Engineering Design Process

Material Selection and Functionality

Multiple cycles of design thinking. Define the problem, explore materials and methods, prototype, test, and refine.

Mapping Process to Design Stages

Engineering design is not linear. Feedback from testing and production continually informs improvements.

Balancing Constraints

Deliberate choice that balances functional requirements, "manufacturability", cost, and sustainability. How engineers translate design goals into real-world material and process decisions.

THE ENGINEERING DESIGN PROCESS

COMMUNICATE

your solution

ITERATE

to improve
your prototype

TEST

and evaluate
your prototype

DEFINE

the problem

IDENTIFY

constraints on your
solution (e.g. time, money,
materials) and criteria
for success

BRAINSTORM

multiple solutions
for the problem

SELECT

the most
promising solution

PROTOTYPE

your solution



Integrated Engineering Design Process

Integrated Engineering Design Process

(multiple disciplines and feedback loops)

Materials + Mechanical + Environmental + Economics



**Integrating engineering design means:
Combining different subjects, ideas, and people with the Engineering
Design Process to solve problems together**

Instead of working in separate steps, this approach encourages teamwork and big-picture thinking to create complete solutions

It helps us build skills in STEM, critical thinking, and practical problem-solving, from the first idea all the way through testing and improving designs

**You see this approach in schools and in big projects
(like designing buildings :)**

System Architecture, Data Pipeline

System Architecture (V-Model Aligned)

Objective: Provide a modular, testable architecture that supports traceability from requirements through validation

High-Level Architecture

- Presentation Layer: Dashboards, APIs, reporting interfaces
- Application / Processing Layer: Business logic, analytics services, orchestration
- Data Layer: Raw, curated, and analytics-ready data stores
- Infrastructure Layer: Compute, storage, networking, security controls

V-Model Mapping

Decomposition & Design (*Left Side*)

System Requirements → System Architecture

Subsystem Design → Component Design

Integration & Verification (*Right Side*)

Component Testing → Subsystem Integration

System Verification → User Acceptance Validation

Configuration Items

Architecture design documents

Interface control documents

Infrastructure-as-Code templates

Versioned service definitions and APIs

End-to-End Data Pipeline

Data Ingestion

- Batch and streaming sources
- Schema validation and data quality checks

Data Processing

- Transformation, aggregation
- Business rule enforcement

Data Storage

- Raw (immutable), curated, and analytics layers

Data Consumption

- BI tools, ML models, downstream services

V-Model Alignment

Requirements → Data definitions, quality rules

Design → Pipeline workflows, storage schemas

Implementation → ETL/ELT jobs, orchestration logic

Verification & Validation:

Data quality tests, Pipeline integration tests, End-user data validation

Key Configuration Items

Data schemas and metadata definitions

Pipeline code (ETL/ELT, orchestration)

Data quality rules and test cases

Versioned datasets and lineage artifacts

Outcome

Full traceability from data requirements to validated outputs

Controlled change management and audit readiness

Project Alignment Questions

1. *V-Model Mapping*: How does your project diagram map requirements → design → implementation (left side) to testing → integration → validation (right side)?
2. *Configuration Items*: What are the key configuration items (e.g., requirements, schemas, code, pipelines, infrastructure), and how are they identified and versioned?
3. *Traceability*: How does the diagram show traceability between requirements, system components, data pipelines, and verification activities?
4. *Verification & Validation*: For each major CI, what test or validation activity is defined and where is it represented on the right side of the V-Model?
5. *Change Impact*: How does the diagram communicate the impact of changes to requirements or data definitions across design, implementation, and testing?

Mini Project Packet

Please be sure to include these answers in your submission

On Canvas - Due 2/16 at 5 PM

<https://canvas.tamu.edu/courses/431435/assignments/2873361>



The Toaster Project

"Hidden in Thomas Thwaites's deceptively simple story is an epic tale that touches on electronics, metallurgy, sustainability, social history, consumerism, epistemology, and global post-industrial capitalism, among other things. It's also funny as hell. Dare I say it? *The Toaster Project* is the twenty-first century's first masterpiece of design writing." —MICHAEL BIERUT, *Pentagram*

"*The Toaster Project* is so cool it is beyond words. It is art and science, history and the future, all in one brilliant idea. Although a tiny project, it is mythic in scope. Once seen, never forgotten." —KEVIN KELLY, *Wired*

"Hello, my name is Thomas Thwaites, and I have made a toaster."

So begins *The Toaster Project*, the author's nine-month-long journey from his local appliance store to remote mines in the UK to his mother's backyard, where he creates a crude foundry. Along the way, he learns that an ordinary toaster is made up of 404 separate parts, that the best way to smelt metal at home is by using a method found in a sixteenth-century treatise, and that plastic is almost impossible to make from scratch. In the end, Thwaites's homemade toaster—a haunting and strangely beautiful object—cost 250 times more than the toaster he bought at the store and involved close to two thousand miles of travel to some of Britain's most isolated locations.

The Toaster Project may seem foolish, even insane. Yet, Thwaites's quixotic tale, told with self-deprecating wit, helps us reflect on the costs and perils of our cheap consumer culture, and in so doing reveals much about the organization of the modern world and our place in it.

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The Toaster Project

Thomas Thwaites

The Toaster Project

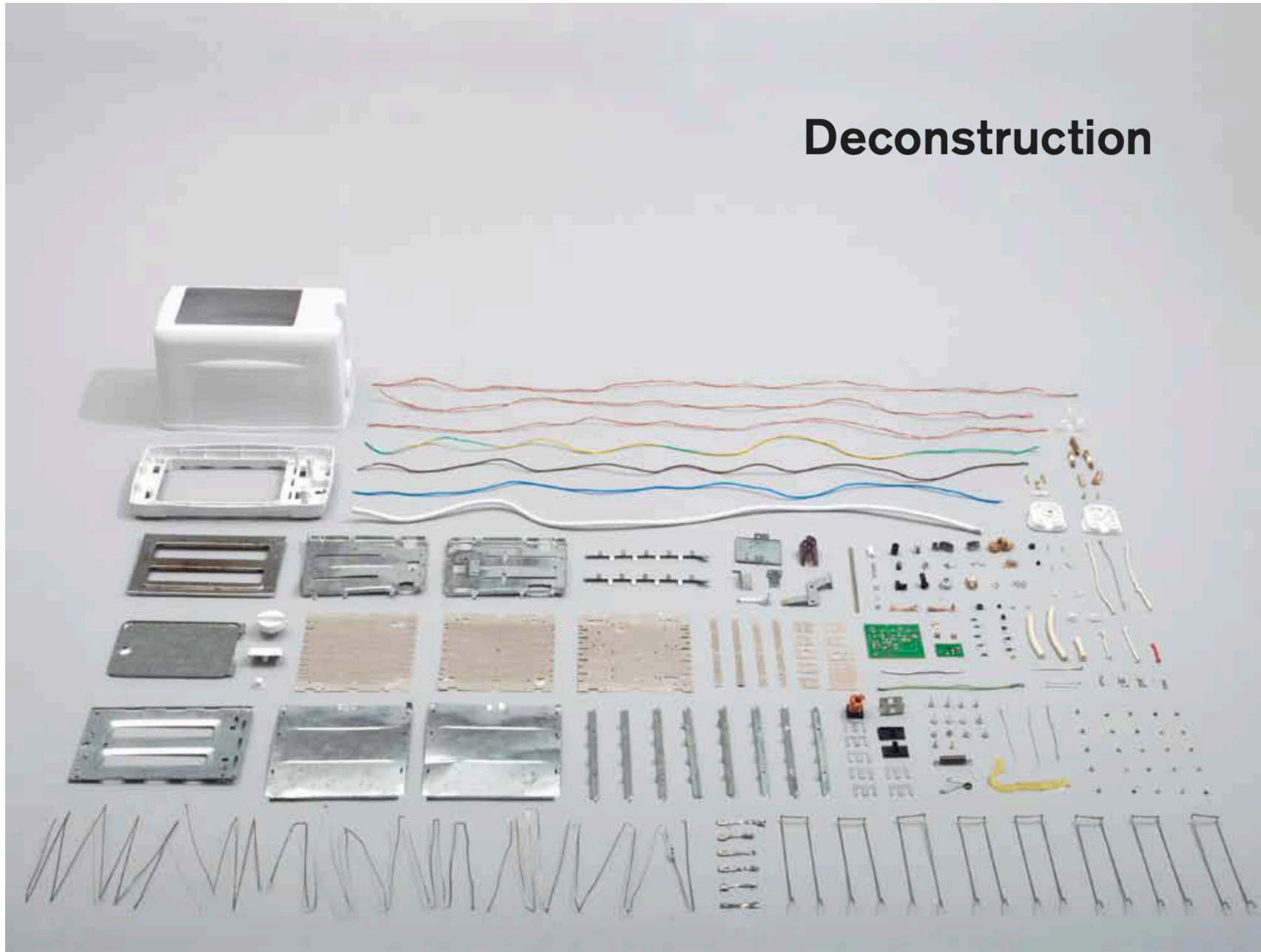
OR A HEROIC ATTEMPT TO
BUILD A SIMPLE ELECTRIC APPLIANCE
FROM SCRATCH



Thomas Thwaites

"Just get married, people give them to you. It's much easier." —Stephen Colbert

Deconstruction





My toaster



Challenges, Solutions

Q1. Feasibility & Hidden Complexity

What assumptions are we making about the simplicity, availability of resources, and manufacturability of our design, and how will we validate or challenge them early?

Expose hidden complexity and prevent unrealistic scope definition.

Q2. Materials, Supply Chains, and Constraints

What critical materials, tools, and external resources does our project depend on, and what are the risks if they are unavailable or delayed?

Early supply-chain awareness and contingency planning.

Q3. Cost and Schedule Realism

How will we estimate cost and schedule realistically, and where do we expect uncertainty, rework, or iteration to occur?

Encourage defensible planning with buffers and explicit risk acknowledgment.

Q4. Performance Definition & Validation

What does “success” mean for our system, and how will we verify and validate performance throughout the project rather than only at the end?

Link requirements to measurable, testable outcomes.

Q5. Process Choices & Tradeoffs

How will we evaluate alternative technical approaches, and what criteria will justify simplifying, changing, or abandoning a design when constraints arise?

Promote pragmatic, evidence-based engineering decisions.

Q6. Documentation & Learning from Failure

How will we document decisions, failures, and lessons learned in real time, and how will this knowledge be used to improve subsequent design choices?

Reinforce engineering as an iterative, learning-driven process.